

ing areas, and transitional aprons or intersections. It is vital for smooth overall safety and efficiency of the airport operations, and sequencing of departure aircrafts. Any aircraft, vehicle, or person walking or working in these areas is required to have 'clearance' from ground control. This is normally done via VHF/UHF radio or using smart phones. Busier airports also use surface movement radar, to display aircrafts and vehicles on the ground, as an additional tool to control ground traffic, especially at night or in poor visibility.

The capabilities of these systems are being enhanced using digital technology. While conventional systems display a map of the airport and the target, latest systems provide high-quality mapping, radar targets, data blocks, safety alerts, and improved interface with other systems.

Air control, also known as 'tower control,' is responsible for the active runway surfaces. It clears aircrafts for takeoff or landing, and ensures that the prescribed runway separation exists at all the time. Under unsafe conditions, a landing aircraft is instructed to 'go-around' and is re-sequenced into the landing pattern. A very efficient and effective communication is maintained between air control and ground control to ensure that the ground control is aware of any operation impacting the taxiways and to create 'gaps' in the arrival traffic to allow taxiing traffic to cross runways and enable departing aircrafts to take off. In advanced systems, applications such as crew resource management and

team resource management make this communication process efficient, reliable and clear.

Clearance and flight data

Clearance delivery system ensures that aircrafts have correct aerodrome information, such as weather and airport conditions, route after de-

recorded continuous loop on a specific frequency, called the automatic terminal information service.

Radar used at the control centres

Control centres use a long-range radar that, at higher altitudes, can see aircrafts within 370km (200 nautical miles) of the radar antenna. These also use TRACON radar to control the areas not covered by the long-range radar.

Since no radar systems are available for traffic control over the oceanic areas, oceanic controllers provide ATC information using procedural control. These procedures use aircraft position reports, time, altitude, distance and speed to ensure separation. The information is recorded on flight progress strips and in specially developed computer systems. Newer systems implement automatic dependent surveillance-broadcast (ADS-B) as part of their surveillance capability.

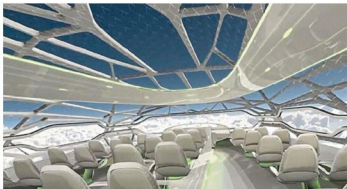
ADS-B reverses the radar concept: Instead of the radar 'finding' a target by interrogating the transponder, the ADS-equipped aircraft sends a position report as determined by the navigation

equipment on board the aircraft. Normally, the aircraft reports a position, automatically or initiated by the pilot, at a predetermined time interval. Controllers may also request more frequent reports to establish aircraft position for specific reasons. ADS can be used where it is not possible to locate a radar system, such as over water.

Computerised radar displays are now being designed to accept



ATC staff using a space-based air traffic surveillance system (Image courtesy: www.satellitetoday.com)



Plane manufacturer Airbus is likely to offer planes with panoramic windows by 2050 (Image courtesy: www.dailymail.co.uk)

parture and time restrictions related to the flight. Busier airports use ground movement planner or traffic management coordinator to prevent taxiway and apron gridlock. Flight data, combined with clearance delivery, ensures that both controllers and pilots have the most current information on weather changes, outages, airport ground delays/ground stops, runway closures, etc. State-of-the-art systems use a